

1.1.3 Constructing interpolants in the Lagrange basis

So, we have improved solving the system of equation from a dense matrix A to an upper triangular A . What if we wanted a diagonal A ?

Lagrange's method uses a special basis of polynomials $\{\ell_j\}$ such that

$$\ell_j(x_i) = \begin{cases} 0 & \text{if } j \neq i \\ 1 & \text{if } j = i \end{cases} \quad (1.3)$$

■ Question 8.

Find the formula for $\ell_j(x)$.

With this basis, the coefficients are just the function values,

$$p_n(x) = \sum_{j=0}^n f(x_j) \ell_j(x) \quad \text{and} \quad \vec{c} = \vec{f}$$

■ Question 9.

Is the Lagrange interpolation polynomial built from a given set of data points same as the Vandermonde interpolation polynomial for the same data?

■ Question 10.

What are the advantages and disadvantages of the Lagrange basis?

■ Question 11.

Lab

Compute the quadratic interpolating polynomial to $f(x) = \cos x$ with nodes $\{-\pi/4, 0, \pi/4\}$ using the Lagrange basis analytically.

Then use your favorite mathematical or programming software to plot the interpolating polynomial and the function in the same picture.

§1.2 Interpolation Error Bounds

Whenever we approximate a function $f(x)$ with a (usually) simpler function, we can create the error function associated with the approximation by taking the difference between the two functions. For instance, the Taylor polynomial of degree n at $x = a$

$$T_n(x) := f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

approximates $f(x)$ with error function equal to the Taylor remainder term

$$f(x) - T_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1},$$

where the unidentified point ξ between x and a can change for different values of x . Now we'll look more carefully at the error involved in f -interpolation.

1.2.1 Errors in f -interpolation

Let's define the error function associated with the f -interpolation 'approximation' of $f(x)$ as

$$E_n(x) = f(x) - p_n(x) \quad (1.4)$$

Since $E_n(x_i) = 0$, let's write

$$E_n(x) = (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n)r(x) = r(x) \prod_{i=0}^n (x - x_i) \quad (1.5)$$

for some "remainder" function $r(x)$.

To determine $r(x)$, we are going to do something that may seem a little strange at this point. We'll define a new function $W(x)$ that is the difference between our two expressions (1.4) and (1.5) for the error function $E_n(x)$, with a slight twist.

Choose a point $\hat{x} \neq x_i$ in $[a, b]$ as the input to the remainder function $r(\hat{x})$, and we leave the other variables x in $E_n(x)$ as they are:

$$W(x) = \underbrace{E_n(x) \text{ via (1.4)}}_{f(x) - p_n(x)} - \underbrace{\text{"twisted" } \hat{E}_n(x) \text{ via (1.5)}}_{(x - x_0)(x - x_1) \cdots (x - x_n)r(\hat{x})}. \quad (1.6)$$

■ Question 12.

□

Given an interval $[a, b]$ containing the set of points $\{\hat{x}, x_0, x_1, \dots, x_n\}$, explain why W has at least $n + 2$ zeroes in $[a, b]$.

■ Question 13.

□

Given an interval $[a, b]$ containing the set of points $\{\hat{x}, x_0, x_1, \dots, x_n\}$, explain why $\exists \xi \in [a, b]$ such that $W^{(n+1)}(\xi) = 0$.

■ Question 14.

□

Use $W(x) = f(x) - p_n(x) - (x - x_0)(x - x_1) \cdots (x - x_n)r(\hat{x})$ to evaluate $W^{(n+1)}(\xi)$. Then simplify and solve $W^{(n+1)}(\xi) = 0$ to find $r(\hat{x})$.

We conclude the following essential result.

Theorem 1.2.4: Interpolation Error Formula (Cauchy)

Suppose $f \in C^{n+1}[a, b]$ and let $p_n \in \mathcal{P}_n$ denote the f -interpolating polynomial with nodes $\{x_0, x_1, \dots, x_n\}$ for distinct points $x_j \in [a, b]$. Then for every $x \in [a, b]$ there exists $\xi \in [a, b]$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{j=0}^n (x - x_j).$$

From this formula we can conclude that the worst error over $[a, b]$ can be estimated by the following bound:

$$\max_{x \in [a, b]} |f(x) - p_n(x)| \leq \left(\max_{\xi \in [a, b]} \frac{|f^{(n+1)}(\xi)|}{(n+1)!} \right) \left(\max_{x \in [a, b]} \prod_{j=0}^n |x - x_j| \right). \quad (1.7)$$

This theorem is the foundation for the convergence theory of piecewise polynomial approximation and interpolatory quadrature rules for definite integrals.

Example 1.2.5

Let's apply the interpolation error bound formula to $f(x) = \sin x$ on $[-5, 5]$. Note that

$$\max_{\xi \in [-5, 5]} |f^{(n+1)}(\xi)| = 1$$

and

$$\max_{x \in [-5, 5]} \prod_{j=0}^n |x - x_j| = 10^{n+1}$$

Hence,

$$\max_{x \in [-5, 5]} |f(x) - p_n(x)| = \frac{10^{n+1}}{(n+1)!}$$

which does $\rightarrow 0$ as $n \rightarrow \infty$.

1.2.2 Conclusions

Here are some conclusions we can draw from [theorem 4](#).

- **“Flatter” $f \implies$ better fit:** The smaller the higher-order derivatives of f are on $[a, b]$, the better the interpolation will fit f . As a special case, any polynomial f will eventually (i.e., for n big enough) be fit exactly by an f -interpolating polynomial.
- **Better fit near center:** The error $E_n(x)$ is smaller if x is centered within $[a, b]$, since that makes the product $(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)$ relatively smaller.
- **Extrapolation is Dangerous:** Using $p_n(x)$ to approximate $f(x)$ for values of x outside the smallest interval containing the original interpolation points $\{x_0, x_1, \dots, x_n\}$ causes relatively larger errors, since the product $(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)$ is relatively larger.

Note: An interesting feature of the interpolation bound is the polynomial $w(x) = (x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)$. This quantity plays an essential role in approximation theory, and also a closely allied subdiscipline of complex analysis called *potential theory*. Naturally, one might wonder what choice of points $\{x_j\}$ minimizes $|w(x)|$ - we will revisit this question in the next section. For now, we simply note that the points that minimize $|w(x)|$ over $[a, b]$ are called *Chebyshev points*, which are clustered more densely at the ends of the interval $[a, b]$.

■ Question 15.

□

Verify that

$$p_1(x) = f(x_0) + \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] (x - x_0)$$

is the unique polynomial of degree ≤ 1 interpolating f at the (distinct) points x_0 and x_1 , and identify what p_1 approaches as the interpolating point x_1 approaches x_0 . Explain your thinking.

■ Question 16.

□

Suppose that p_n is the polynomial of degree $\leq n$ that interpolates $f(x)$ at the distinct points $x_0, x_1, \dots, x_n$. Develop a conjecture about what p_n approaches as the interpolating points $x_1, \dots, x_n$ all approach x_0 simultaneously, and explain your thinking.

Hint: use the definition of $E_n(x)$ and the alternative formula we developed for it in this section.